RELATIVE PSEUDO-COHERENCE OF A COMPLEX OF B-MODULES IS INSENSITIVE TO COMPOSING A -> B WITH A FINITE TYPE R-ALGEBRA MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- If A is pseudo-coherent as a module over R[x] then after the flat base change R[x] -> R[x, y] the algebra A[y] = A \otimes_{R[x]} R[x, y] is pseudo-coherent over R[x, y]. invoked at proof lines 21658-21662. Makes the composed surjection a pseudo-coherent presentation.
- For a surjection A' -> B' of rings with B' pseudo-coherent as an A'-module, m-pseudo-coherence (resp. pseudo-coherence) of a complex of B'-modules over B' is equivalent to m-pseudo-coherence (resp. pseudo-coherence) over A'. invoked at proof lines 21663-21664. Converts the composed presentation into the stated equivalence.

Source reference. Original lemma 066C: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16385–16385. Cited at proof line 21661 to transfer pseudo-coherence of A over R[x] to A[y] over R[x, y]; the conclusion is used inside the counted span.

Source reference. Original lemma 064Z: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16283–16283. Cited at proof line 21664 to convert the composed presentation into the stated equivalence; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $A \to B$ be a map of finite type R-algebras. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of B-modules. Assume A is pseudo-coherent relative to R. Then the following are equivalent*

(1) *$K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) relative to A, and*
(2) *$K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent) relative to R.*

Proof. Choose a surjection $R[x_1, \ldots, x_n] \to A$. Choose a surjection $A[y_1, \ldots, y_m] \to B$. Then we get a surjection

$$R[x_1, \ldots, x_n, y_1, \ldots, y_m] \to A[y_1, \ldots, y_m]$$

which is a flat base change of $R[x_1, \ldots, x_n] \to A$. By assumption A is a pseudo-coherent module over $R[x_1, \ldots, x_n]$ hence by Lemma 066C we see that $A[y_1, \ldots, y_m]$ is pseudo-coherent over $R[x_1, \ldots, x_n, y_1, \ldots, y_m]$. Thus the lemma follows from Lemma 064Z and the definitions. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.